Raj Reddy

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Education

University of Utah **Salt Lake City, UT**
B.S. Computer Science *August 2020 – May 2025*

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies **Dallas, TX**
Associate Software Engineer *June 2025 – Present*

- Owned CI/CD pipeline infrastructure with static analysis, integration testing, and automated validation across multiple production services, reducing deployment friction and maintaining reliability through continuous delivery.
- Built containerized microservices for high-frequency sensor data ingestion using Podman orchestration, engineering for strict latency and throughput constraints in production environments.
- Collaborated cross-functionally with hardware and firmware engineers to define reliability requirements and implement preventive measures, applying post-incident analysis to drive infrastructure improvements.

Doxy.me **Charleston, SC**
Software Engineer *August 2024 – May 2025*

- Deployed and operated production inference services on GCP and AWS using Kubernetes and Fargate, owning observability (structured logging, metrics, distributed tracing), zero-downtime releases, and incident response for systems serving thousands of concurrent clinical users.
- Built and maintained CI/CD pipelines and developer tooling integrating TypeScript and WebRTC services, reducing friction in the engineering workflow and enabling faster, safer delivery across the stack.
- Applied AI-assisted automation to eliminate manual operational toil, building internal tooling that improved engineering productivity and reduced time-to-debug for production incidents.

University of Utah **Salt Lake City, UT**
Undergraduate Research Assistant – FuTURES Lab *May 2024 – Jun 2025*

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

Projects

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

University Course Planning/Review Platform *(Class Best):* Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.